

NYCU Introduction to Machine Learning, Homework 2

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Part. 1, Coding (60%):

(25%) Logistic Regression w/ Gradient Descent Method

1. (5%) Show the hyperparameters (learning rate and iteration, etc) that you used and the weights and intercept of your model.

```
LR = LogisticRegression(  
    learning_rate=1e-3, # You can modify the parameters as you want  
    num_iterations=1000, # You can modify the parameters as you want  
)
```

```
2025-10-23 20:33:35.076 | INFO | __main__:fit:59 - Iteration 1, Train-Acc=0.6506  
2025-10-23 20:33:35.091 | INFO | __main__:main:310 - LR: Weights: [-0.68717698 0.10948854 0.88277671 -0.04912658 0.09460743], Intercep: -2.3612478961151844
```

2. (5%) Show the AUC of the classification results on the testing set.

```
2025-10-23 20:33:35.092 | INFO | __main__:main:311 - LR: Accuracy=0.8333, AUC=0.8523
```

3. (15%) Show the accuracy score of your model on the testing set

```
2025-10-23 20:33:35.092 | INFO | __main__:main:311 - LR: Accuracy=0.8333, AUC=0.8523
```

(25%) Fisher Linear Discriminant, FLD

4. (5%) Show the mean vectors m_i ($i=0, 1$) of each class, the within-class scatter matrix S_w , and the between-class scatter matrix S_b of the training set.

```
2025-10-23 20:33:35.093 | INFO | __main__:main:333 - FLD: m0=[ 0.35994138 -0.04560139], m1=[0.32519126 0.04435118] of cols=['27', '30']  
2025-10-23 20:33:35.094 | INFO | __main__:main:334 - FLD:  
Sw=  
[[41.93041055 15.7202037 ]  
 [15.7202037 37.25186904]]  
2025-10-23 20:33:35.094 | INFO | __main__:main:335 - FLD:  
Sb=  
[[ 0.00120757 -0.00312586]  
 [-0.00312586 0.00809147]]  
2025-10-23 20:33:35.094 | INFO | __main__:main:336 - FLD:
```

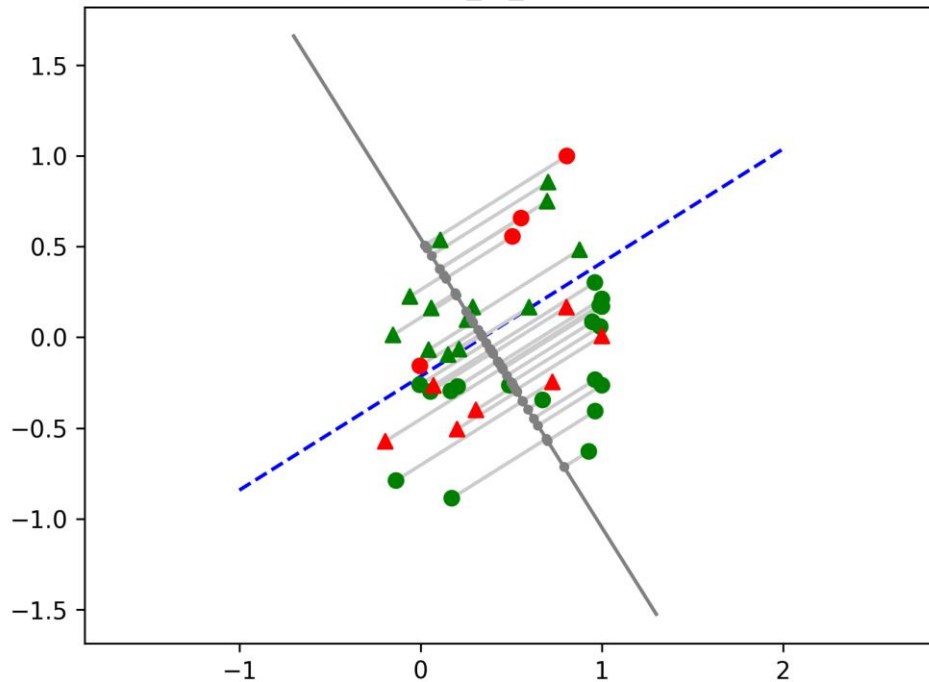
5. (5%) Show the Fisher's linear discriminant w of the training set.

```
2025-10-23 20:33:35.094 | INFO | __main__:main:336 - FLD:  
w=  
[-0.53138255 0.84713198]
```

6. (15%) Show the accuracy score on the testing set. Also, plot/obtain predictions for the testing set by measuring the distance between the projected value of the testing data and the projected means of the training data for the two classes (Please check the slide for color, shape, and other plotting requirements).

```
2025-10-23 20:33:35.094 | INFO | __main__:main:337 - FLD: Accuracy=0.7381
```

Projection onto FLD axis ($w=[-0.53138255 \ 0.84713198]$)
 Slope=-1.594204, Intercept_of_PL=0.545495, Accuracy=0.738095



(10%) Code Check and Verification

- (10%) Lint the code and show the PyTest results.

```
PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\MLHW2\hw2> uv run flake8 .\main.py
PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\MLHW2\hw2>

PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\MLHW2\hw2> uv run flake8 .\main.py
PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\MLHW2\hw2> uv run pytest .\test_main.py -s

===== test session starts =====
platform win32 -- Python 3.12.2, pytest-8.4.2, pluggy-1.6.0
rootdir: C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\MLHW2\hw2
configfile: pyproject.toml
collected 2 items

test_main.py (395, 2) (395,)
2025-10-23 20:38:11.909 | INFO | main:fit:59 - Iteration 1, Train-Acc=0.5360
2025-10-23 20:38:11.922 | INFO | main:fit:59 - Iteration 1001, Train-Acc=0.9960
2025-10-23 20:38:11.933 | INFO | main:fit:59 - Iteration 2001, Train-Acc=1.0000
2025-10-23 20:38:11.942 | INFO | main:fit:59 - Iteration 3001, Train-Acc=1.0000
2025-10-23 20:38:11.957 | INFO | main:fit:59 - Iteration 4001, Train-Acc=1.0000
2025-10-23 20:38:11.957 | INFO | main:fit:59 - Iteration 5001, Train-Acc=1.0000
2025-10-23 20:38:11.978 | INFO | main:fit:59 - Iteration 6001, Train-Acc=1.0000
2025-10-23 20:38:11.993 | INFO | main:fit:59 - Iteration 7001, Train-Acc=1.0000
2025-10-23 20:38:12.005 | INFO | main:fit:59 - Iteration 8001, Train-Acc=1.0000
2025-10-23 20:38:12.007 | INFO | main:fit:59 - Iteration 9001, Train-Acc=1.0000
2025-10-23 20:38:12.028 | INFO | main:fit:59 - Iteration 10001, Train-Acc=1.0000
2025-10-23 20:38:12.040 | INFO | main:fit:59 - Iteration 11001, Train-Acc=1.0000
2025-10-23 20:38:12.047 | INFO | main:fit:59 - Iteration 12001, Train-Acc=1.0000
2025-10-23 20:38:12.063 | INFO | main:fit:59 - Iteration 13001, Train-Acc=1.0000
2025-10-23 20:38:12.075 | INFO | main:fit:59 - Iteration 14001, Train-Acc=1.0000
2025-10-23 20:38:12.080 | INFO | main:fit:59 - Iteration 15001, Train-Acc=1.0000
2025-10-23 20:38:12.099 | INFO | main:fit:59 - Iteration 16001, Train-Acc=1.0000
2025-10-23 20:38:12.112 | INFO | main:fit:59 - Iteration 17001, Train-Acc=1.0000
2025-10-23 20:38:12.126 | INFO | main:fit:59 - Iteration 18001, Train-Acc=1.0000
2025-10-23 20:38:12.130 | INFO | main:fit:59 - Iteration 19001, Train-Acc=1.0000
2025-10-23 20:38:12.147 | INFO | main:fit:59 - Iteration 20001, Train-Acc=1.0000
2025-10-23 20:38:12.161 | INFO | main:fit:59 - Iteration 21001, Train-Acc=1.0000
2025-10-23 20:38:12.174 | INFO | main:fit:59 - Iteration 22001, Train-Acc=1.0000
2025-10-23 20:38:12.183 | INFO | main:fit:59 - Iteration 23001, Train-Acc=1.0000
2025-10-23 20:38:12.193 | INFO | main:fit:59 - Iteration 24001, Train-Acc=1.0000
2025-10-23 20:38:12.210 | INFO | main:fit:59 - Iteration 25001, Train-Acc=1.0000
```

```

2025-10-23 20:38:12.223 | INFO | main:fit:59 - Iteration 26001, Train-Acc=1.0000
2025-10-23 20:38:12.232 | INFO | main:fit:59 - Iteration 27001, Train-Acc=1.0000
2025-10-23 20:38:12.242 | INFO | main:fit:59 - Iteration 28001, Train-Acc=1.0000
2025-10-23 20:38:12.257 | INFO | main:fit:59 - Iteration 29001, Train-Acc=1.0000
2025-10-23 20:38:12.274 | INFO | test_main:test_logistic_regression:35 - accuracy=1.0000
. (395, 2) (395,)
2025-10-23 20:38:12.274 | INFO | test_main:test_fld:45 - accuracy=0.8759
.
===== 2 passed in 2.07s =====
PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\MLHW2\hw2>

```

Part. 2, Questions (40%):

1. (15%)

(*) Using (4.57) and (4.58), derive the result (4.65) for the posterior class probability in the two-class generative model with Gaussian densities, and verify the results (4.66) and (4.67) for the parameters $\mathbf{w}$ and w_0 .

$$\begin{aligned}
 p(\mathcal{C}_1|\mathbf{x}) &= \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \\
 &= \frac{1}{1 + \exp(-a)} = \sigma(a) \quad (4.57)
 \end{aligned}$$

$$a = \ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \quad (4.58)$$

$$p(\mathcal{C}_1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + w_0) \quad (4.65)$$

$$\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \quad (4.66)$$

$$w_0 = -\frac{1}{2}\boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{p(\mathcal{C}_1)}{p(\mathcal{C}_2)}. \quad (4.67)$$

Write your answer here.

By Handouts Ch3 page 41.

$$p(\mathbf{x}|\mathcal{C}_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_k) \right\}$$

$$p(x | \mathcal{C}_1) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp \left[-\frac{1}{2}(x - \mu_1)^T \Sigma^{-1}(x - \mu_1) \right]$$

$$p(x | \mathcal{C}_2) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp \left[-\frac{1}{2}(x - \mu_2)^T \Sigma^{-1}(x - \mu_2) \right]$$

Start from

$$a = \ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \quad (4.58)$$

$$a = \ln \left(\frac{p(x | \mathcal{C}_1)p(\mathcal{C}_1)}{p(x | \mathcal{C}_2)p(\mathcal{C}_2)} \right)$$

$$\begin{aligned}
&= \ln \left(\frac{p(C_1)}{p(C_2)} \exp \left\{ -\frac{1}{2}(x - \mu_1)^T \Sigma^{-1}(x - \mu_1) - \left[-\frac{1}{2}(x - \mu_2)^T \Sigma^{-1}(x - \mu_2) \right] \right\} \right) \\
&= \ln \frac{p(C_1)}{p(C_2)} + \left[-\frac{1}{2}(x - \mu_1)^T \Sigma^{-1}(x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma^{-1}(x - \mu_2) \right] \\
&\longrightarrow \left[-\frac{1}{2}(x - \mu_1)^T \Sigma^{-1}(x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma^{-1}(x - \mu_2) \right] \\
&= \frac{1}{2} \left[\left(-\mathbf{x}^T \Sigma^{-1} \mathbf{x} + \mu_1^T \Sigma^{-1} \mathbf{x} + \mathbf{x}^T \Sigma^{-1} \mu_1 - \mu_1^T \Sigma^{-1} \mu_1 \right) \right. \\
&\quad \left. + \left(\mathbf{x}^T \Sigma^{-1} \mathbf{x} - \mu_2^T \Sigma^{-1} \mathbf{x} - \mathbf{x}^T \Sigma^{-1} \mu_2 + \mu_2^T \Sigma^{-1} \mu_2 \right) \right]
\end{aligned}$$

Because

$$a^T A b = b^T A^T a$$

And

$$(\Sigma^{-1})^T = \Sigma^{-1}$$

So

$$\begin{aligned}
\mu_1^T \Sigma^{-1} \mathbf{x} + \mathbf{x}^T \Sigma^{-1} \mu_1 &= 2\mu_1^T \Sigma^{-1} \mathbf{x} \\
-\mu_2^T \Sigma^{-1} \mathbf{x} - \mathbf{x}^T \Sigma^{-1} \mu_2 &= -2\mu_2^T \Sigma^{-1} \mathbf{x}
\end{aligned}$$

So

$$\begin{aligned}
&= \frac{1}{2} \left[\left(-\mathbf{x}^T \Sigma^{-1} \mathbf{x} + \mu_1^T \Sigma^{-1} \mathbf{x} + \mathbf{x}^T \Sigma^{-1} \mu_1 - \mu_1^T \Sigma^{-1} \mu_1 \right) \right. \\
&\quad \left. + \left(\mathbf{x}^T \Sigma^{-1} \mathbf{x} - \mu_2^T \Sigma^{-1} \mathbf{x} - \mathbf{x}^T \Sigma^{-1} \mu_2 + \mu_2^T \Sigma^{-1} \mu_2 \right) \right] \\
&= \frac{1}{2} \left[\left(2\mu_1^T \Sigma^{-1} \mathbf{x} - 2\mu_2^T \Sigma^{-1} \mathbf{x} + \mu_2^T \Sigma^{-1} \mu_2 - \mu_1^T \Sigma^{-1} \mu_1 \right) \right] \\
&= \left[(\mu_1^T - \mu_2^T) \Sigma^{-1} \mathbf{x} \right] + \frac{1}{2} (\mu_2^T \Sigma^{-1} \mu_2 - \mu_1^T \Sigma^{-1} \mu_1)
\end{aligned}$$

So

$$\begin{aligned}
a &= \ln \frac{p(C_1)}{p(C_2)} + \left[-\frac{1}{2}(x - \mu_1)^T \Sigma^{-1}(x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma^{-1}(x - \mu_2) \right] \\
&= \left[(\mu_1^T - \mu_2^T) \Sigma^{-1} \mathbf{x} \right] + \frac{1}{2} (\mu_2^T \Sigma^{-1} \mu_2 - \mu_1^T \Sigma^{-1} \mu_1) + \ln \frac{p(C_1)}{p(C_2)} \\
&= \mathbf{w}^T \mathbf{x} + w_0
\end{aligned}$$

By

$$\begin{aligned}
p(\mathcal{C}_1|\mathbf{x}) &= \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \\
&= \frac{1}{1 + \exp(-a)} = \sigma(a) \\
p(\mathcal{C}_1 | x) &= \sigma(a) = \sigma(\mathbf{w}^T x + w_0)
\end{aligned} \tag{4.57}$$

Because

$$\mathbf{a}^T \mathbf{A} \mathbf{b} = \mathbf{b}^T \mathbf{A}^T \mathbf{a}$$

And

$$(\Sigma^{-1})^T = \Sigma^{-1}$$

So

$$\begin{aligned}
\mathbf{w}^T &= (\mu_1^T - \mu_2^T) \Sigma^{-1} \\
\Rightarrow \mathbf{w} &= \Sigma^{-1} (\mu_1 - \mu_2)
\end{aligned}$$

Finally prove

$$p(\mathcal{C}_1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + w_0) \tag{4.65}$$

$$\mathbf{w} = \Sigma^{-1}(\mu_1 - \mu_2) \tag{4.66}$$

$$w_0 = -\frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_2^T \Sigma^{-1} \mu_2 + \ln \frac{p(\mathcal{C}_1)}{p(\mathcal{C}_2)}. \tag{4.67}$$

2. (10%)

(a) Give one real-world situation where you would prefer Logistic Regression (LR) over the Perceptron, and explain why.

(b) Is Logistic Regression actually used for regression (predicting a continuous value)? If not, state what task it really solves and why the name includes “regression.”

Write your answer here.

(a) Medical diagnosis (For example, judging tumor is benign or malignant). LR give “Probability” , which can be seem as how confidence it is; Perception only knows whether the result is benign or malignant. In this situation, it is better for the doctors to judge the final result. If doctors know the probability, they can decide and adjust next medical treatment to be more intense or not, but if they only know the class of results, it is difficult or impossible for them to do that.

(b) No. Logistic Regression is used to solve classification problems (usually binary), the name includes "Regression" because it uses a structure like "Linear Regression."

Linear Regression:

$$y = w^T x + w_0$$

Logistic Regression:

$$p = \sigma(w^T x + w_0)$$

It computes a ($z = w^T x + w_0$) like linear regression to update it and finally uses $\sigma()$ [Sigmoid function] to let z to become a probability p between (0, 1).

3. (15%)

(a) Why is feature scaling (e.g., standardization or normalization) important in Logistic Regression? Explain two reasons.

(b) If feature scaling is not applied in Logistic Regression, list three problems that may occur. Briefly explain.

Write your answer here.

(a) Reason 1: Accelerates Convergence and Stabilizes the Optimization Process

If the feature scales are very different (for example, one is 0 – 1, and another is 0 – 10,000), the contour plot of the loss function appears as an "ellipse", when gradient descent searches for the lowest point, the update would oscillate zigzagging between the narrow walls. Standardization or normalization can bring all features to a similar scale, making gradient descent more stable and faster.

Reason 2: Prevent Certain Features Dominating

If some features have a particularly large scale, they influence the loss function more, models tend to adjust their weights, while other features may be ignored. The weight reflects the "scale", rather than the "importance" of feature. Standardization or normalization brings all features to a similar scale, making all features compete fairly.

(b) Which some features may have a particularly large scale.

Problem 1: gradient descent converges slowly and unstably, and updates would oscillate zigzagging.

Problem 2: certain features may dominate, weight reflects the "scale", rather than the "importance".

Problem 3: ineffective regularization, the penalty is unfairly concentrated on small scale weights (which have large scale weight).